

# Bridging Graph Polynomials and Symmetric Tensors

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14 July 2026

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# Overview

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  - 2 Investigate establishing a Belkale-Brosnan type result for symmetric tensors

# Symmetric Tensor Recap

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- Elementary symmetric tensors are of the form  $v^{\otimes d}$ .

# Relationship with Homogenous Polynomials

Consider  $k[x_1 \dots x_n]_d$ , the set of all homogenous polynomials of degree  $d$  over  $k$  (implicitly included in this is the 0 polynomial). Then an established result is that

$$k[x_1 \dots x_n]_d \cong S^d(k^n)$$

as  $k$ -vector spaces, by linearly extending the mapping of monomials

$$\mathbf{x}^\alpha := \prod_{i=1}^n x_i^{a_i} \mapsto \frac{1}{d!} \sum_{\sigma \in S_d} \sigma \left( \bigotimes_{i=1}^n e_i^{\otimes a_i} \right)$$

Here  $\alpha$  denotes the tuple  $(a_1, a_2, \dots, a_n)$  where  $a_i \geq 0 \forall i = 1, 2, \dots, n$ , and for this mapping  $|\alpha| = \sum_{i=1}^n a_i = d$ .

# Veronese Variety

Geometrically, consider  $\mathbb{P}V$  to be the set of all lines in  $V \cong k^n$  passing through the origin. Then the Veronese mapping given by

$$\nu_d : \mathbb{P}V \rightarrow \mathbb{P}k^N$$

taking points

$$[x_1 : x_2 : \dots : x_n] \mapsto [x_1^d : x_1^{d-1}x_2 : \dots : x_n^d]$$

allows for the visualisation of elementary (rank-1) symmetric tensors as points on a variety in  $\mathbb{P}k^N$ . Here  $N = \binom{n-1+d}{d}$ . More general symmetric tensors then lie on secant varieties to the Veronese.

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- A *tree* is a connected graph with no loops i.e.  
 $b_0(G) = 1, b_1(G) = 0$ .
- A *spanning tree*  $T \leq G$  contains all of  $V$  but not (necessarily) all of  $E$ .

# Constructing (particular) Graph Polynomials

For the rest of this presentation  $G = (V, E)$  such that  $\|G\| = n$  and  $b_0(G) = 1$ . A *Kirchoff* polynomial for a given graph  $G$  is given by

$$P_G(x_1, \dots, x_n) := \sum_{T \leq G} \prod_{e \notin T} x_e$$

Note the variables of the polynomial correspond to a choice of labelling the edges.

# Properties of the Kirchoff Polynomial

- They are homogenous polynomials, and lie in the quotient ring  $R := k[x_1, x_2, \dots, x_n]/(x_1^2, x_2^2, \dots, x_n^2)$

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- Variables common to each term in the polynomial denote self-loops of  $G$ .
- An equivalence relation  $\sim_S$  can be imposed by  $P_A \sim_S P_B \iff P_A(x_1, \dots, x_n) = \tau \cdot P_B(x_1, \dots, x_n)$ . Here  $\tau \in S_n$  permutes  $n$  elements, corresponding to a relabelling of edges.



# What is... the Grothendieck Ring of Varieties?

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 $[X] = \{V \mid \pi : X \dashrightarrow V \text{ a birational morphism exists}\}$ ; modulo all relations of the form  $[X] = [X \setminus Y] + [Y]$ . Addition here is defined as  $[X] + [Y] = [X \sqcup Y]$

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- This group can be seen as a commutative ring with unity by defining multiplication  $[X][Y] = [X \times_k Y]$ .
- $\mathcal{K}(V_k)$  can be seen as a module over  $\mathbb{Z}[\mathbb{L}]$  where  $\mathbb{L} := [\mathbb{A}^1]$

# Belkale, Brosnan (2004)

Let  $\mathbb{V}(P_G) \subset \mathbb{A}^n$  denote the scheme of zeroes of  $P_G$ . Then define  $Y_G = \mathbb{A}^n \setminus \mathbb{V}(P_G)$ . Then

Theorem (Belkale, Brosnan 2004)

*Define  $S := \{\mathbb{L}^n - \mathbb{L}\}_{n \geq 2}$  as a multiplicative subset in  $\mathcal{K}(V_k)$ . Then the collection of  $[Y_G]$  generate  $S^{-1}\mathcal{K}(V_k)$  as a  $S^{-1}\mathbb{Z}[\mathbb{L}]$  module.*

## Aluffi, Marcolli 2011

$X, Y$  are stably birational iff  $X \times \mathbb{P}k^m \dashrightarrow Y \times \mathbb{P}k^n$  is birational, for  $n, m \geq 0$ .

Theorem (Larsen, Lunts (2001))

$\mathcal{K}(V_k)/(\mathbb{L}) \cong \mathbb{Z}[SB]$ , the ring of stably birational varieties

Theorem (Aluffi, Marcolli (2011))

The images of  $[Y_G]$  in  $\mathcal{K}(V_k)$  generate  $\mathbb{Z} \subsetneq \mathbb{Z}[SB]$

Corollary

Suppose  $S' := \{\mathbb{L}^n - 1\}_{n \geq 1}$ . Then  $[Y_G]$  do not generate  $S'^{-1}\mathcal{K}(V_k)$  over  $S'^{-1}\mathbb{Z}[\mathbb{L}]$

# Graphs as a Disjoint Union

Let  $\mathcal{G}_n$  denote the family of all connected graphs with a fixed number of edges  $n$ .

- The corresponding graph polynomials all lie in  $R := k[x_1, \dots, x_n]/(x_1^2, \dots, x_n^2)$ .

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- Now  $\mathbf{G}_n := (\mathcal{G}_n / \sim_{KP}) / \sim_S$  partitions the space of finite graphs into classes having the same Kirchoff polynomial upto relabelling.
- $\mathbf{G}_n$  can then be expressed as the disjoint union  $\bigsqcup_{i=0}^n \mathbf{G}_{n,i}$  where each component contains the graphs with  $b_1(G) = i$ .

# Addition in Graphs?

In  $\mathbf{G}_n$ , an attempt to define an addition operation is given below

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- Choosing  $P_C$  as an element with the largest number of summands in this set.  $[P_C]_S$  then is a corresponding equivalence class.
- **Conjecture:** Does there always exist  $C$  a graph such that its Kirchoff polynomial is  $P_C$ ? If yes then
- This turns  $\mathbf{G}_n$  into an additive abelian group



# Image of $R$ in symmetric tensors

*What does  $R$  (the ring of Kirchoff polynomials) look like when using  $\Phi$  the isomorphism between homogenous polynomials and symmetric tensors? (**Note:**  $k$  is no longer required to be  $\mathbb{F}_2$ )*

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- However, since none of the elements in  $R$  can be expressed as linear polynomials to the  $d$ -th power, they cannot be seen as lying on the Veronese variety.
- Consider the mapping  $\mu_d : \mathbb{P}V \rightarrow \mathbb{P}k^{(n)}_d$  taking  $[x_1 : \dots : x_n] \mapsto [\prod_{i=1}^d x_i : \dots : \prod_{i=n-d+1}^n x_i]$ .

# Kirchoff Polynomials as Symmetric Tensors

$$\begin{array}{ccc}
 \mathbb{P}V & \xrightarrow{\nu_d} & \mathbb{P}k^{\binom{n+d-1}{d}} \\
 & \searrow \mu_d & \nearrow i \\
 & \mathbb{P}k^{\binom{n}{d}} &
 \end{array}$$

Thus Kirchoff polynomials can be seen as specific points on  $\sigma_r(i(\mu_d(\mathbb{P}V))) \subseteq \sigma_r(\nu_d(\mathbb{P}V))$  where  $r$  denotes the rank of the image tensor under  $\Phi$ .

# Symmetric Tensors and the Grothendieck Ring

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- Every symmetric tensor can be associated therefore to a variety (and its complement)
- In particular the symmetric tensors associated to graph polynomials then correspond to  $[Y_G]$  as defined earlier as well as the complement  $\mathbb{V}(P_G)$ .

# Generation of Symmetric Tensors

*What does the element  $[\mathbb{A}^1] =: \mathbb{L} \in \mathcal{K}(V_k)$  correspond to when working with symmetric tensors or homogenous polynomials?*

# Further Questions

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- What is the significance of the image  $\mathbb{L}$  when working with symmetric tensors?
- Does applying a grading on the space of graphs only work when dealing with the  $\mathbb{F}_2$ ? Can this be extended to any field  $k$ ?

# Bibliography I